

GRADE 6 SCIENCE · ALBERTA CURRICULUM · MATTER

Matter

Guided Notes Package

Name: _____

Date started: _____

Before We Begin: Pre-Knowledge Check*No marks. No correction. Be honest.*

Write everything you think you know about matter and particles. What is matter made of? What happens to matter when it heats up or cools down? What is the difference between a solid, a liquid, and a gas?

You'll return to this page at the end of the unit. Keep it honest — seeing how your thinking changed is part of the learning.

Section 1: The Particle Model

Learning target: I can explain what the particle model says, describe how particles are arranged and moving, and predict what happens to particles when matter is heated or cooled.

VOCABULARY

Term	Definition in your own words	One example
Matter		
Particle		
Particle model		
Temperature		
Kinetic energy		

NOTES

What is matter?

Matter is anything that has _____ and takes up _____. The chair you sit on, the air you breathe, the water you drink — all of these are matter.

The particle model — a scientific abstraction

Scientists use a model called the **particle model of matter** to explain how matter behaves. The particle model has four key ideas:

1. All matter is made up of tiny _____ that are too small to see with the naked eye.
2. There are _____ between particles — particles are not packed together with zero space.
3. Particles are always _____. They never stop completely.
4. The _____ of that movement changes with temperature.

Connection to Computer Science: The particle model is an **abstraction** — it keeps only what matters (particle spacing and movement speed) and removes everything else (actual molecular structure, chemical bonding, quantum behaviour). In computer science, abstraction means making a simplified version of something complex that is still useful for the job. A map is an abstraction of a city — it removes buildings, textures, and smells, but keeps streets and directions. The particle model does the same thing for matter.

Write one other example of an abstraction you use in everyday life:

What temperature actually measures:

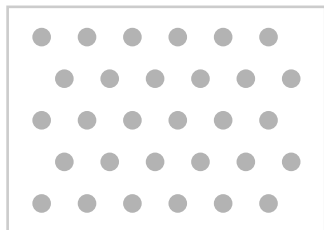
Temperature is a measure of how _____ particles are moving on average. It is not a measure of heat stored — it is a measure of particle motion.

When matter gets _____ (temperature increases), particles move faster and spread further apart.

When matter gets _____ (temperature decreases), particles move slower and pull closer together.

Particle diagrams — what particles look like at different temperatures:

COLD (low temperature)



Slow, closer together

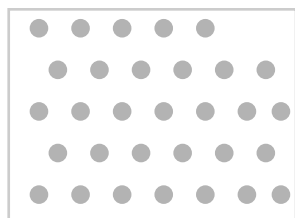
HOT (high temperature)



Fast, further apart

Label the boxes below: write “hot” or “cold” and “faster” or “slower” under each.

Diagram A



Temperature: _____
Particles moving: _____

Diagram B



Temperature: _____
Particles moving: _____

“A particle is just a tiny piece of the material — like a tiny piece of steel.”

This is one of the most common misunderstandings in this unit. A particle in the particle model is **not** a miniature version of the material. Particles are discrete units with **empty space between them** — that space is real, and it is what allows matter to expand and contract. If particles were stuck together with no space, nothing could ever change size with temperature. The space between particles is as important as the particles themselves.

Summarising the particle model:

What the model says	Fill in the blank
All matter is made of	_____
Particles have _____ between them	space
Particles are always	_____
Heating makes particles move	_____ and spread _____
Cooling makes particles move	_____ and come _____

Retrieval Check — Close your notes

1. Name two things the particle model says are always true about particles, no matter what the temperature is.

2. What does temperature actually measure, according to the particle model?

3. What happens to the space between particles when matter is heated?

4. A student heats a metal rod. Based on the particle model, explain what is happening to the particles inside the rod.

5. ★★ A classmate says: “The particle model is wrong because you can’t see particles — so how do we know they’re real?” How would you respond? Use at least one piece of evidence or reasoning to support the particle model.

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Now — go to the workbook. You can now answer **4 questions** in the workbook. Close your notes and go answer them.

Section 2: States of Matter and Phase Changes

Learning target: I can describe the particle arrangement and movement in solids, liquids, and gases, name the six phase changes, and explain why mass is conserved during a phase change even when volume changes.

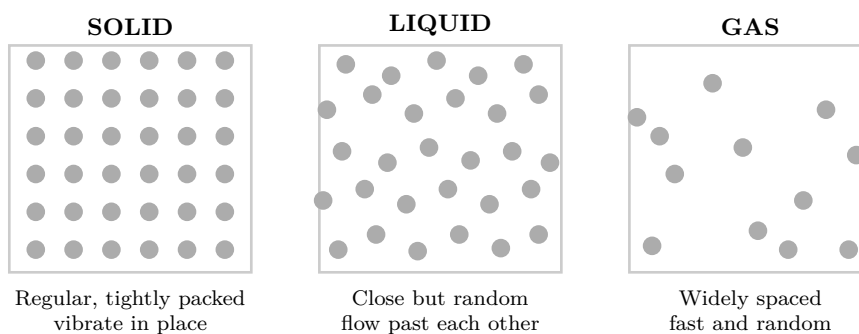
VOCABULARY

Term	Definition in your own words	One example
Solid		
Liquid		
Gas		
Phase change		
Melting		
Freezing		
Evaporation		
Condensation		

NOTES

The three states of matter

The particle model explains the three states of matter by describing how particles are arranged and how fast they move.



Solid:

- Particles are packed _____ together in a regular arrangement
- Particles _____ (vibrate) but do not move past each other
- The shape is _____ — solids hold their shape without a container
- The volume is _____

Liquid:

- Particles are _____ together but NOT in a fixed arrangement

- Particles can slide _____ each other — they flow
- The shape is _____ — liquids take the shape of their container
- The volume is _____ (a liquid doesn't compress easily)

Gas:

- Particles are _____ apart — they fill all available space
- Particles move _____ and in random directions
- The shape is _____ — gases fill their container completely
- The volume is _____ — gases can be compressed

Comparison table — fill in the blanks:

Property	Solid	Liquid	Gas
Particle spacing	_____	close	_____
Particle movement	vibrate in place	_____ past each other	fast and random
Fixed shape?	_____	No	No
Fixed volume?	Yes	_____	_____

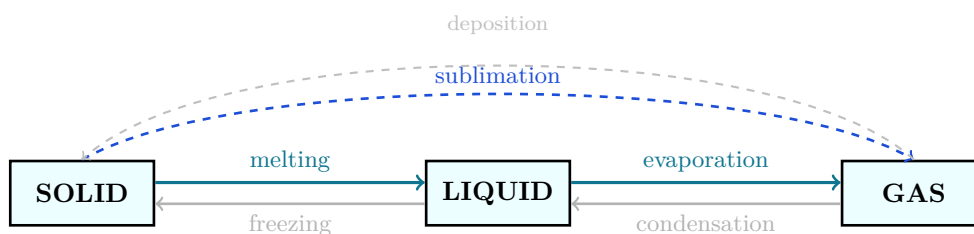
Phase changes — moving between states

A **phase change** happens when matter changes from one state to another. The temperature changes, so the speed of particle movement changes — enough to change how particles are arranged.

The six phase changes:

Change	Direction	Name
Solid → Liquid	Heating	_____
Liquid → Solid	Cooling	_____
Liquid → Gas	Heating	_____
Gas → Liquid	Cooling	_____
Solid → Gas	Heating (rapid)	Sublimation
Gas → Solid	Cooling (rapid)	Deposition

Phase change diagram — circle the changes that involve HEATING; box the changes that involve COOLING:

**The experimental logic of phase changes:**

Scientists who study phase changes control one variable at a time. To test how temperature affects the phase change of water, you would:

- Keep the same _____ of water each time

- Change only the _____ (or the heat source)
- Observe the temperature at which the _____ happens

CRITICAL IDEA — Mass is conserved during a phase change:

When matter changes state, _____ is conserved. Only the _____ and _____ of the particles change.

100 g of ice → melts → still _____ g of liquid water

100 g of liquid water → evaporates → still _____ g of water vapour

The mass does NOT change. Volume, however, _____ change during a phase change.

“When ice melts, some of the mass disappears — that’s why there’s less water.”

Mass is **never** created or destroyed during a phase change. The ice and the puddle of water weigh exactly the same. What changes is the **arrangement** of particles and the **volume** they take up — not the number of particles or their mass. If water seems to “disappear” when it evaporates, it has not vanished — it has become water vapour in the air, still carrying the same mass. Conservation of mass is one of the most fundamental laws in science.

Think about a time you saw a phase change in your everyday life. Describe what you saw and name the phase change. Use the particle model to explain what was happening to the particles.

Retrieval Check — Close your notes

1. Describe the particle arrangement in a solid versus a gas. Use the words “spacing” and “movement” in your answer.

2. Name the phase change that occurs when: (a) liquid water becomes ice, and (b) liquid water becomes water vapour.

(a) _____ (b) _____

3. A student heats 50 g of ice until it is completely melted. What is the mass of the liquid water?

Explain why, using the particle model.

4. Why does a gas fill the entire container it is placed in, while a solid keeps its own shape?

5. ★★ A student claims that boiling water loses mass because “the steam floats away.” Write a response that corrects this using the conservation of mass and the particle model. What would you need to measure to prove the student wrong?

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Now — go to the workbook. You can now answer **4 questions** in the workbook. Close your notes and go answer them.

Section 3: Thermometers and Temperature

Learning target: I can explain how a thermometer works using the particle model, read the Celsius scale, and identify key reference points and safety practices.

VOCABULARY

Term	Definition in your own words	One example
Celsius scale		
Reference point		
Thermal expansion		
Calibration		

NOTES

The Celsius scale

Temperature is measured in degrees Celsius ($^{\circ}\text{C}$) in science and in everyday life in Canada.

The Celsius scale uses two reference points based on water:

Reference point	Temperature
Water _____ (liquid $\rightarrow$ solid)	0°C
Water _____ (liquid $\rightarrow$ gas)	100°C
Normal human body temperature	$\approx 37^{\circ}\text{C}$
Room temperature (comfortable)	$\approx 20\text{--}22^{\circ}\text{C}$
A cold winter day in Alberta	$\approx -20^{\circ}\text{C}$

These reference points were chosen because they are _____ (they always happen at the same temperature, at standard conditions) and easy to reproduce in any lab.

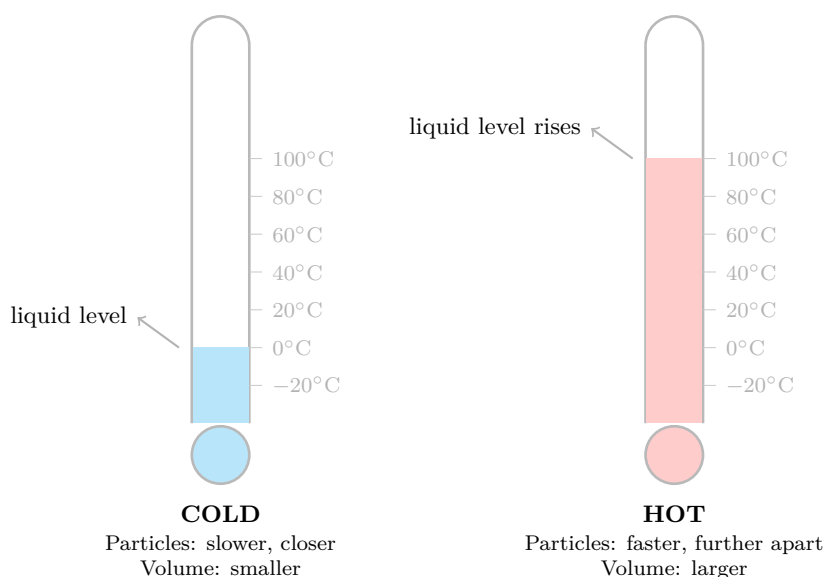
How a thermometer works

A traditional liquid thermometer works because of the same particle behaviour you just studied:

1. The thermometer contains a liquid (often coloured alcohol) sealed in a glass tube.
2. When temperature _____, the liquid particles move faster and spread further apart. The liquid _____ and rises up the tube.
3. When temperature _____, the liquid particles slow down and come closer together. The liquid _____ and falls down the tube.
4. A scale on the glass shows what temperature the liquid level matches.

The thermometer works because the _____ of the liquid changes predictably with temperature.

Thermometer at two temperatures:



Safety practices — thermometers in the lab

- Never _____ the thermometer — hold it gently; it can crack under pressure
- Never use a thermometer to _____ liquids — it is not a stirring rod
- If a thermometer _____, do not touch the glass or liquid — notify your teacher
- Read the thermometer at _____ — your eye should be level with the liquid surface (parallax error if read from above or below)
- Wait for the reading to _____ before recording — the liquid needs time to reach the temperature of the substance

Digital thermometers do not use liquid expansion. They use a tiny electrical sensor that changes resistance with temperature. The reading on the screen is still in Celsius. What do you think stays the same between a liquid thermometer and a digital thermometer, and what is different?

Retrieval Check — Close your notes

1. What are the two reference points on the Celsius scale? Give the temperature for each.

2. Explain, using the particle model, why the liquid in a thermometer rises when temperature increases.

3. A student reads a thermometer from above the liquid level. What kind of error might this cause?

4. A student uses the thermometer to stir a beaker of hot water. What could go wrong, and why?

5. ★★ The Celsius scale is described as “calibrated to water.” What does this mean, and why does using water as the reference make the scale useful for both everyday life and science experiments involving living things?

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Now — go to the workbook. You can now answer **3 questions** in the workbook. Close your notes and go answer them.

Section 4: Expansion, Contraction, and the Water Anomaly

Learning target: I can explain why most materials expand when heated and contract when cooled, describe real-world applications, explain why water is an exception, and use the particle model to explain why ice floats.

VOCABULARY

Term	Definition in your own words	One example
Expansion		
Contraction		
Density		
Anomaly		
Expansion joint		

NOTES

The general rule: expansion and contraction

- When a material is _____, its particles move faster and spread further apart. The material takes up more space — it _____.
- When a material is _____, its particles slow down and move closer together. The material takes up less space — it _____.

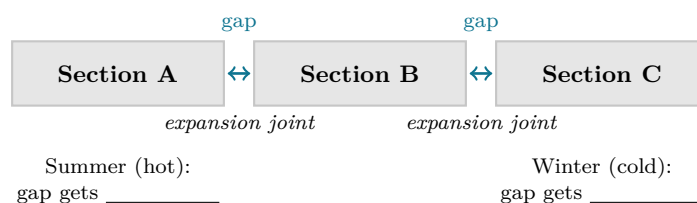
This applies to solids, liquids, and gases — though gases expand and contract far more dramatically.

Real-world applications:

Engineers must account for expansion and contraction in design. Materials fixed rigidly in place that cannot expand will _____, crack, or buckle.

Expansion joints in bridges and roads:

Long bridges are not one continuous piece. Engineers leave small _____ between sections — called expansion joints.



Without expansion joints, the force of expansion would _____ the bridge.

Gaps in railway tracks: Traditional rail sections also leave small gaps for the same reason. Modern continuous welded rail is pre-stressed to account for _____.

Power lines: Electrical lines hang lower in _____ (heat causes sagging) and are pulled tighter in _____ (cold causes contraction).

Look at the road or a bridge near your school. Why would a road in Alberta need larger expansion joints than a road built in a tropical region with stable temperatures year-round?

The water anomaly — when the general rule breaks

For almost every material, the solid is _____ than the liquid (tightly packed = smaller volume for same mass = higher density).

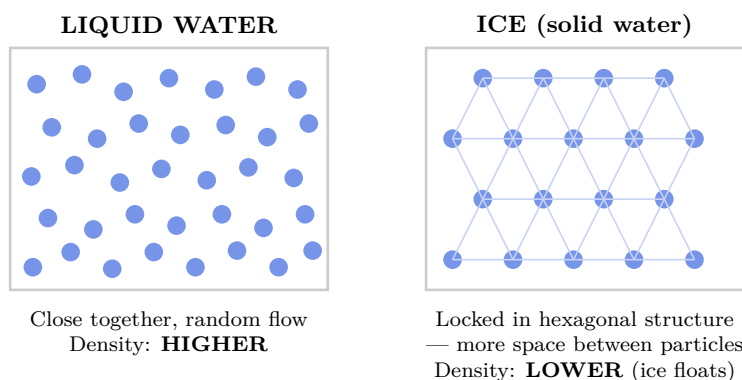
Water breaks this rule. This is called the **water anomaly**. When water freezes:

- Most liquids become more _____ when they solidify
- Water becomes _____ dense when it freezes

Ice takes up _____ space than the same mass of liquid water. Ice is less dense than liquid water. This is why ice _____.

Why water is different — the particle model explanation:

When water freezes, particles lock into a crystal structure with a _____ arrangement that actually pushes particles *further apart* — creating more space, not less.



“Ice is denser than liquid water — solids are always denser than liquids.”

For most materials, this is true — but **water is a major exception**. Ice is **less** dense than liquid water. This is why ice floats at the surface rather than sinking to the bottom. The reason is the structure water forms when it freezes: a hexagonal crystal lattice that takes up **more** space than randomly arranged liquid water particles. This is not a small exception — it has enormous consequences for life on Earth.

Why the water anomaly matters:

Aquatic life in winter: When a lake cools, ice forms on the _____ and floats. The liquid water beneath stays at approximately _____ — cold, but not frozen. Fish and other organisms survive in the liquid layer _____. If ice sank, lakes could freeze _____ to _____ and most aquatic life would not survive.

Infrastructure: Water that seeps into rock cracks and freezes expands, breaking rock apart — called freeze-thaw weathering. Engineers designing roads in cold climates must account for this in drainage

design.

FNMI Knowledge

Cree, Dené, and Métis communities in northern Alberta have observed and documented seasonal freeze-thaw cycles on rivers and lakes for generations. This knowledge includes when ice is thick enough for travel, how ice changes near river currents, and how early melt or late freeze signals broader ecological shifts. This is a form of long-term data collection — systematic observation across seasons and years — that aligns with what scientists now measure using sensors and satellite data. Both are ways of tracking the same real patterns.

The water anomaly as a testable prediction:

Most people would predict ice sinks. The particle model, if incorrectly applied, predicts: solid is denser, so solid sinks. The observation — ice floats — **surprises** that prediction. This is exactly how science works: when a result surprises your prediction, you do not discard the observation. You *revise your model*. The hexagonal crystal structure explanation came *after* scientists observed that ice floats and had to figure out why.

Summarising expansion, contraction, and the water anomaly:

Situation	What happens	Why
Metal rod heated	_____	Particles move faster, spread apart
Metal rod cooled	_____	Particles slow down, come closer
Water heated (liquid)	expands slightly	Particle spacing increases
Water cooled to 0°C and frozen	_____ (anomaly)	Crystal structure pushes particles _____

Retrieval Check — *Close your notes*

- What is the general rule for how most materials behave when heated?

- What is an expansion joint, and why do bridges need them?

- Ice floats on liquid water. What does this tell you about the density of ice compared to liquid water?

- Why can fish survive in an ice-covered lake in winter? Use the words “density” and “layer” in your answer.

- ★★ A student says: “The fact that ice floats proves the particle model is wrong, because solids are supposed to be denser.” Is this a correct critique of the particle model? Explain why the water anomaly does not disprove the model — and what it actually shows about

how science works when observations surprise predictions.

Self-check: ☐ I could teach it ☐ Getting there ☐ Not yet

Now — go to the workbook. You can now answer **5 questions** in the workbook. Close your notes and go answer them.

Unit Synthesis — From Memory

Close your notes. Reconstruct.

For each section, write the single most important idea in one sentence — from memory, no looking back.

Section 1 — The Particle Model

Section 2 — States of Matter and Phase Changes

Section 3 — Thermometers and Temperature

Section 4 — Expansion, Contraction, and the Water Anomaly

Now open your notes and check.

Guiding question for this unit: How can the particles of matter be influenced by heating or cooling?

Write your answer in 3–5 sentences from memory. Use at least three science vocabulary terms from this unit.

Final step — return to your Pre-Knowledge Check. Read what you wrote before the unit began.

What did you get right? _____

What were you wrong about? _____

What surprised you most? _____

Self-Assessment Rubric

Rate yourself honestly after the final session. Use this to identify what to review before the unit test.

Knowledge cluster	I can explain this without notes			One question I still have
Particle model: particles have space, always move, speed changes with temperature	Yes yet	Getting there	Not yet	
Three states of matter: particle arrangement and movement in solid, liquid, gas	Yes yet	Getting there	Not yet	
Phase changes: names, directions, and conservation of mass	Yes yet	Getting there	Not yet	
Thermometers: how they work, Celsius scale, reference points, safety	Yes yet	Getting there	Not yet	
Expansion and contraction: general rule, real-world applications	Yes yet	Getting there	Not yet	
Water anomaly: ice is less dense, floats, why this matters for aquatic life	Yes yet	Getting there	Not yet	

Bring your unanswered questions to class. They will appear on the inquiry project and the unit test.